IT5437 Assignment 2 — Fitting and Alignment

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Q1(a) — Total Least Squares on Line 1

Mean-centre (x_1, y_1), form A. SVD(A): right singular vector of *smallest* singular value is the line normal — minimising perpendicular (total) distances, not vertical residuals.

```
_, _, Vt = np.linalg.svd(A)
a, b = Vt[-1]; c = a*mx + b*my
slope = -a/b; intercept = c/b
```

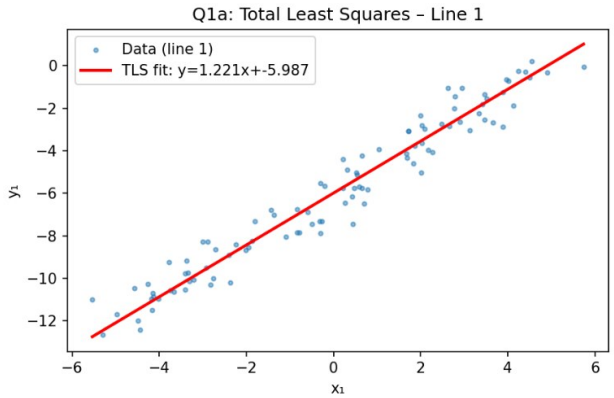
Result: y = 1.2207 x – 5.9872 (normal: 0.7736x – 0.6337y = 3.7942)

Q1(b) — RANSAC: Three Lines from All 300 Points

Flatten all three x/y columns into one 300-pt cloud. RANSAC (2 000 iterations, perpendicular-distance threshold 0.5): find best consensus, mask its inliers, repeat ×3.

```
for i in range(3):
    a,b,c,inl = ransac_line(pts, thresh=0.5)   # perp-dist TLS
    remaining = np.setdiff1d(remaining, remaining[inl])
```

Lines: y=–0.454x+2.087 (84 pts) | y=1.280x–6.001 (66 pts) | y=1.032x+1.084 (66 pts)



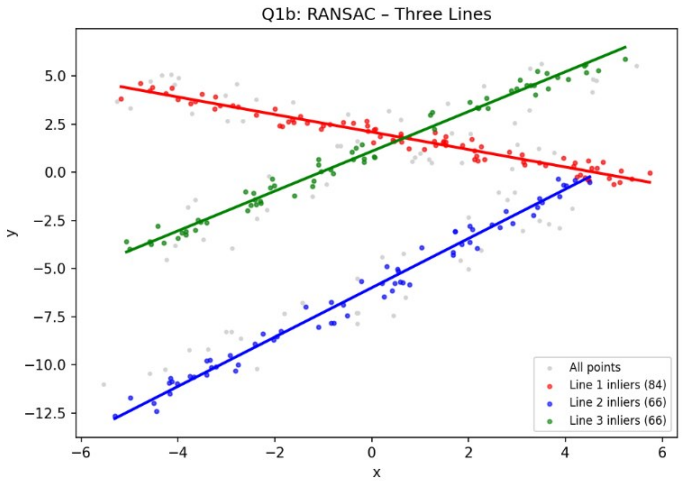


Fig.1 (left): TLS fit — Line 1 data only. Fig.2 (right): Sequential RANSAC — three lines from the combined cloud.

Q2 — Real-World Earring Size (Pinhole Camera)

f = 8 mm, pixel pitch p = 2.2 µm = 0.0022 mm, distance d = 720 mm. real_size (mm) = N_{px} × p × d / f.

```
real_w = pixel_w * 0.0022 * 720 / 8   # e.g. 365 px → 72.3 mm
```

Threshold (gray < 200) isolates earrings from the white background. Bounding boxes:

Earring	W (px)	H (px)	W (mm)	H (mm)
1	365	394	72.3	78.0
2	358	394	70.9	78.0

Each earring ≈ 71–72 mm wide × 78 mm tall — a standard large hoop (~7–8 cm diameter).

Fig.3. Left: earring image. Right: detected bounding boxes (blue).

Q3(a) — Manual Homography: Warp c1 → c2

6 landmark pairs identified visually (DC jack, USB connector, MCU chip top-left/bottom-right, red reset LED, top-right corner board edge). cv2.findHomography (no RANSAC) gives H from exactly 6 pairs.

```
pts_src = [(219,222),(319,129),(197,365),(326,341),(234,478),(399,484)]
pts_dst = [(254,176),(376,113),(195,308),(326,319),(201,427),(359,476)]
H, _ = cv2.findHomography(np.float32(pts_src), np.float32(pts_dst))
```

H encodes a ≈**15° rotation + translation** — consistent with two photos of the same board from slightly different viewpoints.

Q3(b) — Difference Image (Manual H)

diff = cv2.absdiff(warped, c2). Mean |diff| = **18.3/255**. Residuals appear at board edges and fine pin labels.

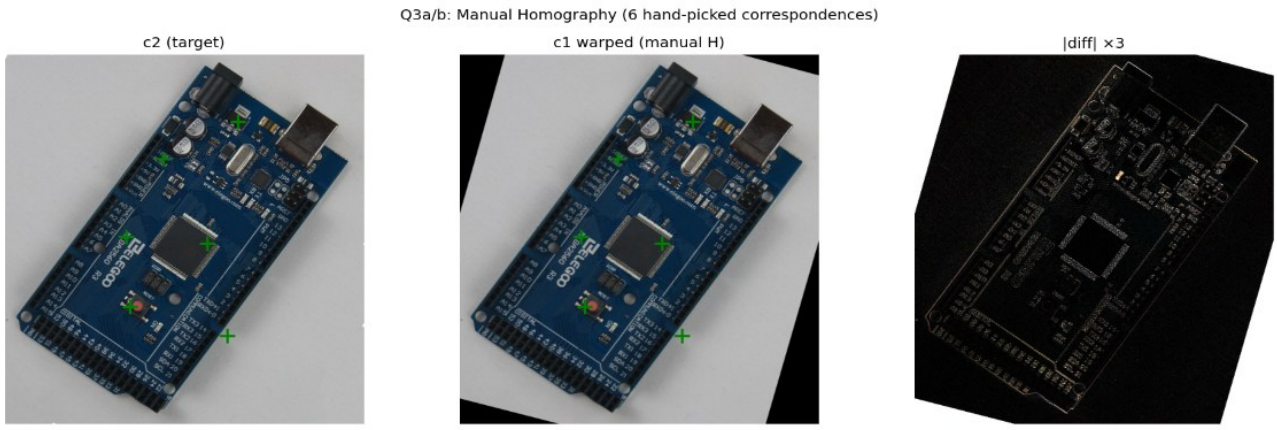


Fig.4. Left: c2 target (+ landmarks). Centre: c1 warped by manual H. Right: |diff|×3.

Q3(c) — SIFT Feature Detection and Matching

1 358 keypoints in c1, 1 464 in c2. BFMatcher (L2) + Lowe ratio test (0.75) → **901 reliable matches**.

```
sift = cv2.SIFT_create()
kp1,des1 = sift.detectAndCompute(gray1,None)
good = [m for m,n in bf.knnMatch(des1,des2,k=2) if m.distance<0.75*n.distance]
```

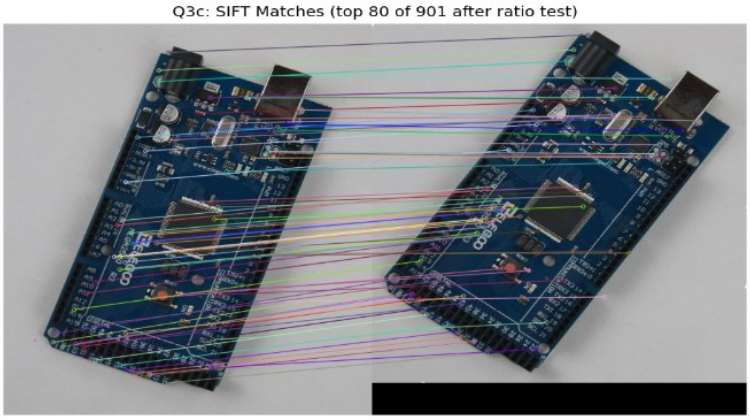


Fig.5. Top 80 of 901 SIFT matches. Green lines connect corresponding keypoints.

Q3(d) — Auto Homography from SIFT Matches (RANSAC)

901 matches → cv2.findHomography (RANSAC, threshold 4 px). **861/901 inliers**; H re-estimated from all inliers. Mean |diff| = **17.1/255**.

```
H_auto, mask = cv2.findHomography(src_pts, dst_pts, cv2.RANSAC, 4.0)
warped = cv2.warpPerspective(im1, H_auto, (w2, h2))
```

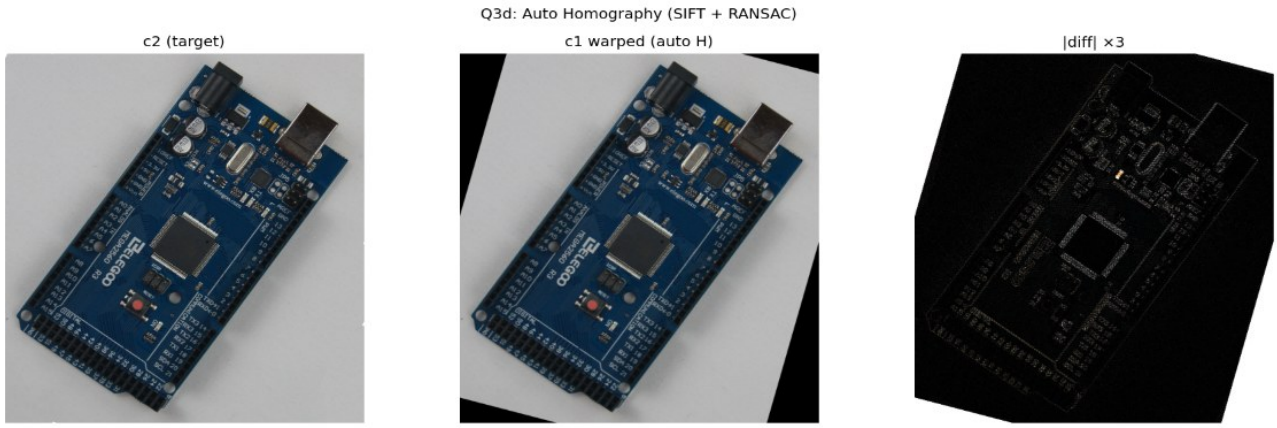


Fig.6. Left: c2 target. Centre: c1 warped by auto H. Right: |diff|×3.

Discussion — Manual vs. Automatic Homography

Accuracy: auto H gives mean |diff|=17.1 vs. 18.3 for manual — a 7% improvement. Both matrices represent essentially the same ≈15° rotation + translation, confirming both methods are correct.

Why auto is more accurate: 861 well-distributed inliers vs. only 6 landmarks. The overdetermined least-squares fit from hundreds of matches is far more robust than 6 hand-picked points, whose localisation is imprecise. RANSAC also automatically discards 40 outlier matches.

Residuals in both methods arise from: (i) photometric differences between shots (lighting angle), (ii) minor out-of-plane board warp not modelled by a planar homography, and (iii) JPEG compression artefacts.